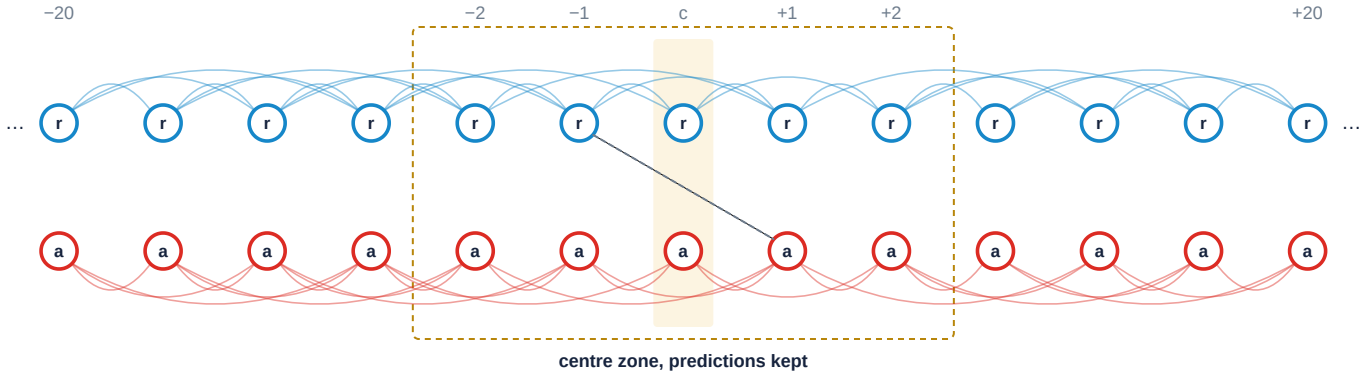
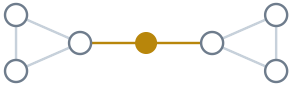
a Window around a trigger variant (PE ≥ 0.8)



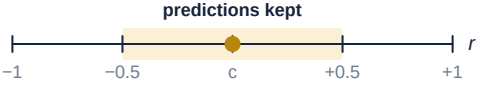
hap-1 allele edges hap-2 allele edges cross-haplotype edge ≤ 41 variants, ≤ 82 allele nodes

b Bridge-vertex feature



removing this variant disconnects the window
→ **is_bridge = 1**

c Centre-zone mask



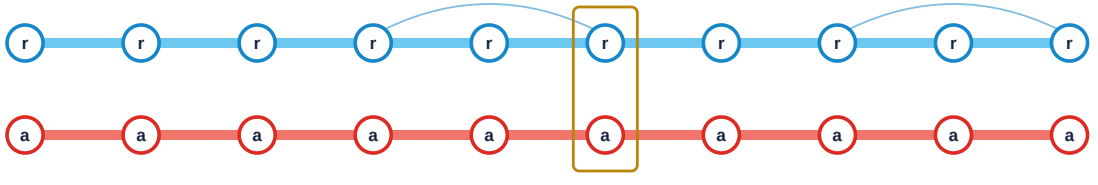
$r = (pos - pos_c) / \max|offset|;\ |r| \le 0.5$

d Exported phasing graph (DOT)

```
1001.1 -> 1543.1 [label=16.0] 1001.2 -> 1543.2 [label=1.1]
```

position.allele → position.allele, label = vote weight cast at the source

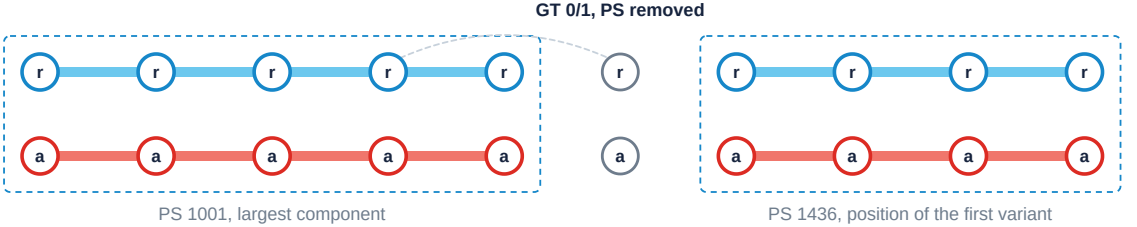
e Before correction: one phase set



PS 1001, GT 0|1 / 1|0

score = 0.62 ≥ 0.35

f After correction: variant unphased, phase set split



edge no longer counted

genotypes are never flipped